

Managing the Unmanageable: UDCA Strategies in Fatty Liver Disease

01

Fatty Liver

Fatty Liver: dari “NAFLD” ke “Steatotic Liver Disease (SLD)”

Steatotic Liver Disease (SLD) = spektrum penyakit hati dengan steatosis (lemak di hepatosit) akibat berbagai etiologi

Subklasifikasi utama SLD:

- MASLD (metabolic dysfunction–associated steatotic liver disease): steatosis + disfungsi metabolik
- MASH (metabolic dysfunction–associated steatohepatitis): MASLD + inflamasi/ballooning (biasanya konfirmasi histologi)
- MetALD: steatosis + disfungsi metabolik + konsumsi alkohol di atas batas MASLD namun tidak setinggi ALD berat

Metabolic dysfunction-associated steatotic liver disease (MASLD), previously termed non-alcoholic fatty liver disease (NAFLD), is defined as steatotic liver disease (SLD) in the presence of one or more cardiometabolic risk factor(s) and the absence of harmful alcohol intake. The spectrum of MASLD includes steatosis, metabolic dysfunction-associated steatohepatitis (MASH, previously NASH), fibrosis, cirrhosis and MASH-related hepatocellular carcinoma (HCC). This joint EASL–EASD–EASO guideline provides an update on definitions, prevention, screening, diagnosis and treatment for MASLD. Case-finding strategies for MASLD with liver

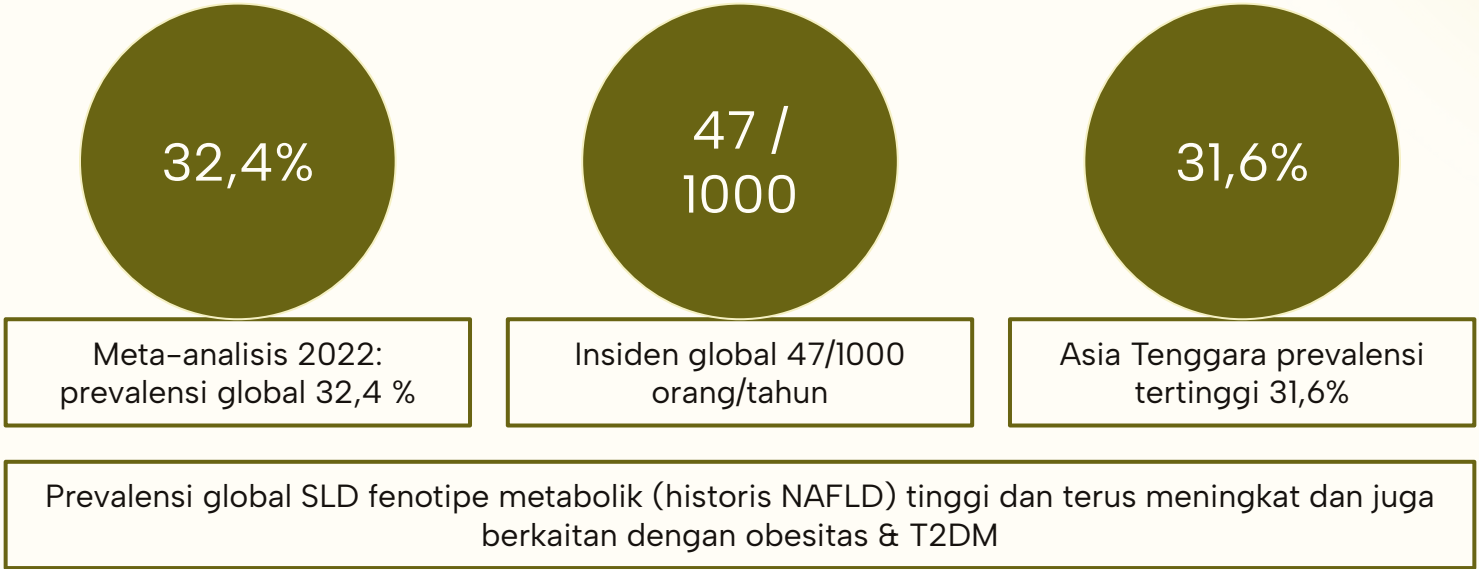
Definisi praktis MASLD/MASH

MASLD ditegakkan bila ada:

Bukti steatosis (USG/CT/MRI/CAP/FibroScan atau biomarker/score), dan ≥ 1 faktor disfungsi metabolik (misal obesitas/overweight, T2DM, atau komponen sindrom metabolik)

MASH adalah subset MASLD dengan hepatoseluler ballooning + inflamasi (umumnya histology-proven)

Epidemiologi Fatty Liver Global



32,4%

Meta-analisis 2022:
prevalensi global 32,4 %

47 /
1000

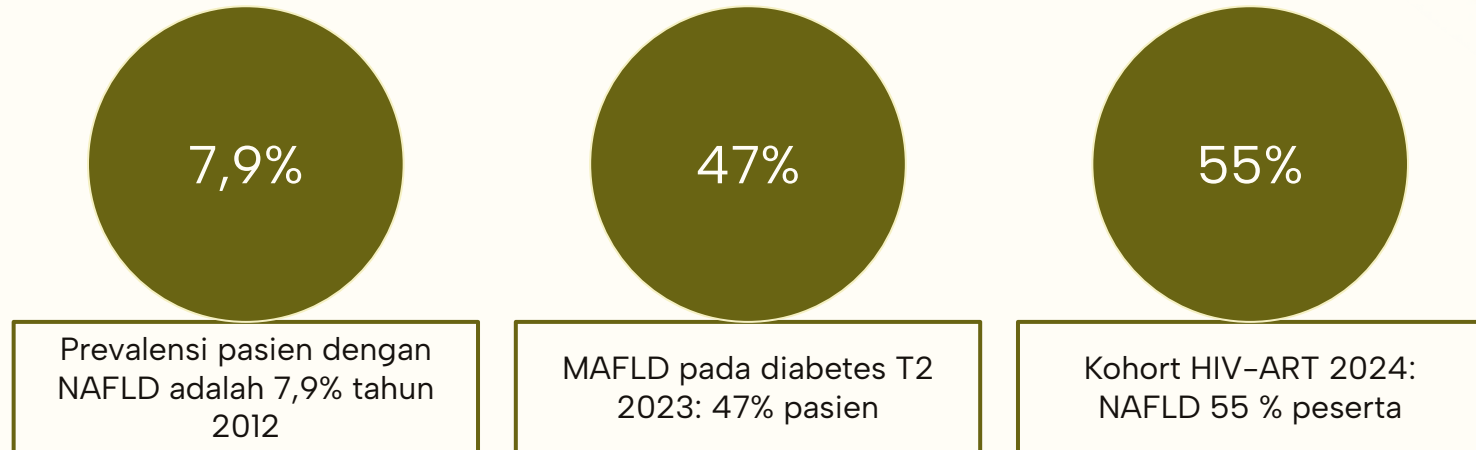
Insiden global 47/1000
orang/tahun

31,6%

Asia Tenggara prevalensi
tertinggi 31,6%

Prevalensi global SLD fenotipe metabolik (historis NAFLD) tinggi dan terus meningkat dan juga berkaitan dengan obesitas & T2DM

Epidemiologi Fatty Liver di Indonesia



Faktor Risiko Fatty Liver

Sindrom metabolik jadi kriteria diagnosis MAFLD

Obesitas viseral

Resistensi insulin

Diabetes T2

Dislipidemia

Table 2 Risk factors for MAFLD

Modifiable risk factors	Non-modifiable risk factors
Metabolic syndrome	Older age
Overweight/obesity	Male sex
Central obesity	Genetic variations (<i>PNPLA3</i> , <i>TM6SF2</i> , <i>GCKR</i> , <i>MBOAT7</i> , <i>MERTK</i> and <i>HSD17B13</i>)
Type 2 diabetes mellitus	
Dyslipidemia	
Insulin resistance	
Hypertension	
High-caloric/fructose diet	
Low level of physical activity	
Sarcopenia	
Myosteatosis	
Gut dysbiosis	
Obstructive sleep apnea	
Hypothyroidism	

Notably, many of these factors could be association, it is hard to ascertain the causality
PNPLA3 patatin-like phospholipase domain-containing protein 3, *TM6SF2* transmembrane 6 superfamily member 2, *GCKR* glucokinase regulator, *MBOAT7* membrane Bound O-Acyltransferase Domain Containing 7, *HSD17B13* hydroxysteroid 17-beta dehydrogenase-13

Comorbidity SLD

Comorbidity	Assessment/parameter
Obesity	BMI, Waist circumference, Waist to height ratio, TSH and free thyroxine (if suspicion of hypothyroidism)
Type 2 diabetes or insulin resistance	Fasting plasma glucose, HbA1c, Oral glucose tolerance test, 2 h post-load glucose, Fasting plasma insulin and/or C-peptide, HOMA-IR
Dyslipidaemia	Fasting plasma triglycerides, Fasting plasma total, LDL- and HDL-cholesterol, Once in a lifetime: measurement of lipoprotein (a)Further investigations ^a :• Non-esterified fatty acids• Apolipoprotein B
Kidney disease	Creatinine in plasma and urine Albumin in serum and urine Estimated glomerular filtration rate (eGFR)
Cardiovascular disease	Fasting plasma uric acid Serum high-sensitivity C-reactive protein (hsCRP) Serum ferritin Systolic and diastolic blood pressure
Atherosclerosis	Complete blood count, Platelets, Elevated lipoprotein
Obstructive sleep apnoea	Neck circumference, Epworth score
PCOS	LH, FSH, testosterone, SHBG, Ovarian ultrasound

Prevalensi Faktor Risiko di Indonesia

50.1%

Proporsi obesitas
dewasa meningkat
dari 23.0% 2007 →
50.1% 2018

21,7%

Prevalensi Metabolik
Sindrom di Indonesia
2018: 21,7%

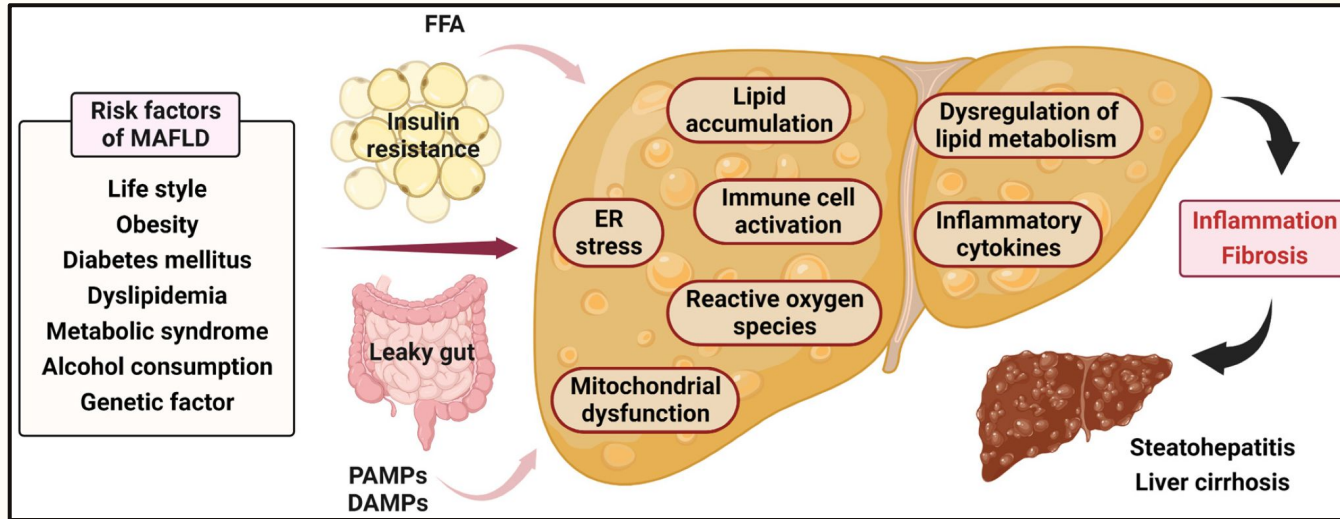
11,3%

Federasi Diabetes
Internasional (IDF)
memperkirakan
prevalensi diabetes
di Indonesia 2024:
11,3%

15,5%

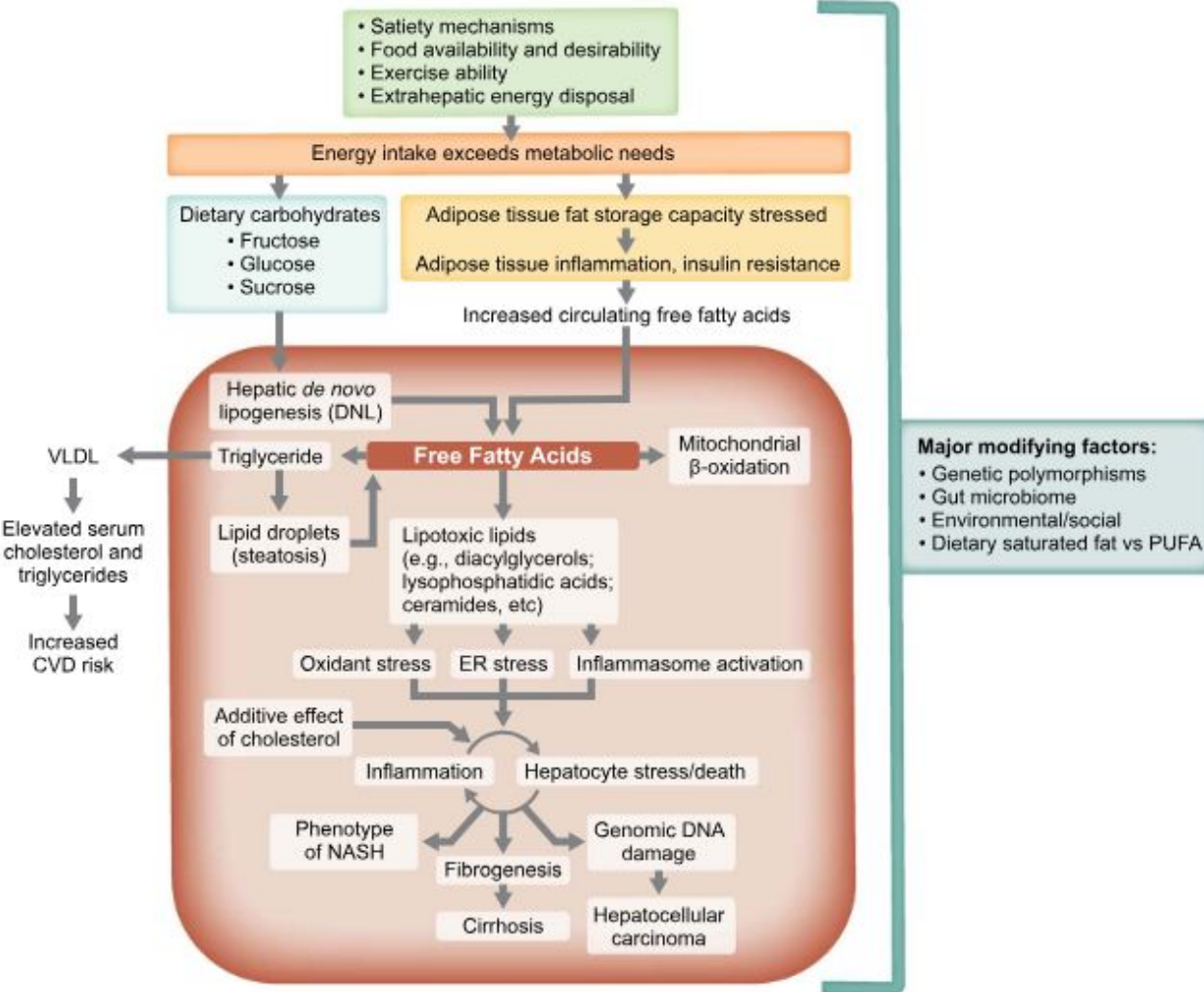
Prevalensi
Dislipidemia di
Indonesia: 15,5%

Patofisiologi Fatty Liver



- Faktor risiko metabolik memicu resistensi insulin → FFA meningkat.
- FFA memasuki hepatosit, menumpuk sebagai lipid droplets (steatosis).
- Lipid overload menimbulkan ER stress & mitochondrial dysfunction, menghasilkan ROS.
- ROS serta PAMPs/DAMPs (dari usus bocor) mengaktifkan TLR-4/9 & inflammasom NLRP3, memicu sitokin pro-inflamasi.
- Disregulasi metabolisme lipid serta lipotoksin memperparah cedera sel.
- Sel stellata teraktivasi → fibrosis; siklus inflamasi–fibrosis menciptakan peradangan kronik.
- Akhirnya, steatosis berkembang menjadi steatohepatitis lalu sirosis hati.

Patofisiologi Fatty Liver



Energi intake & karbohidrat berlebih → hati mengubahnya menjadi trigliserida via DNL

Jaringan lemak kelebihan muatan membocorkan FFA dan menimbulkan resistensi insulin

FFA masuk hepatosit, menumpuk sebagai lipid droplets dan lipotoksin

Lipotoksin memicu ROS, ER-stress, disfungsi mitokondria → kerusakan sel

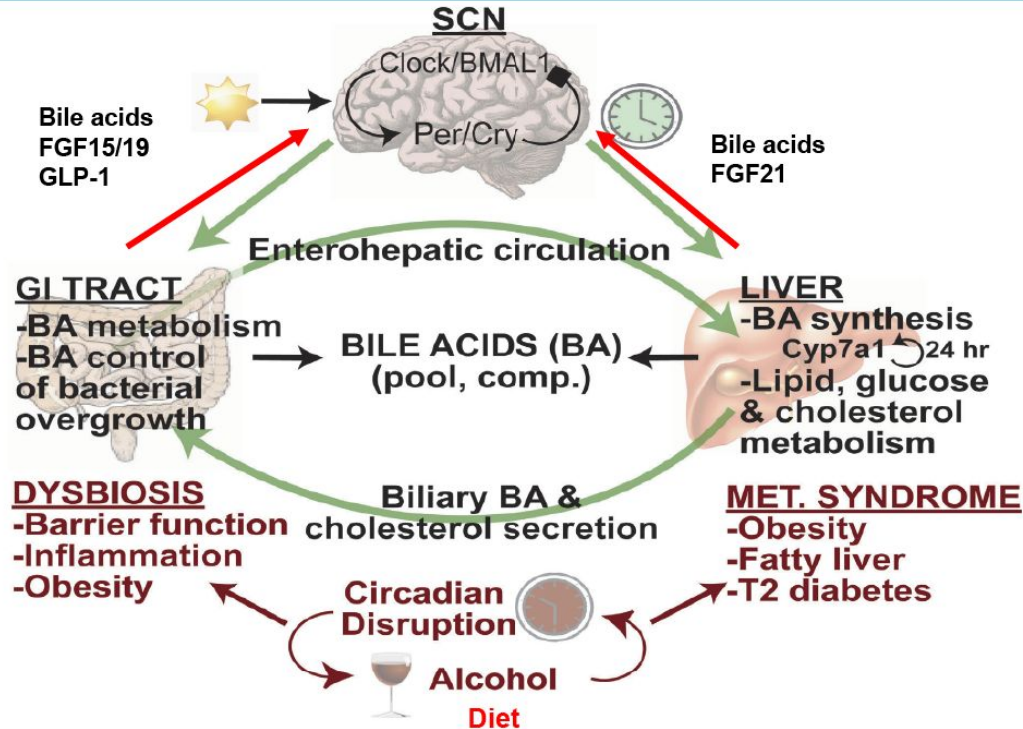
Kerusakan mengaktifkan mediator inflamasi & sitokin → peradangan kronik memperparah cedera

Stellate cell teraktivasi → fibrosis → sirosis → HCC

Genetik, jenis lemak diet, mikrobiota usus, hipoksia OSA, alkohol & lingkungan mengubah kecepatan perubahan

Hati berlemak meningkatkan VLDL & trigliserida → risiko kardiovaskular ↑↑

Bile Acid & Fatty Liver



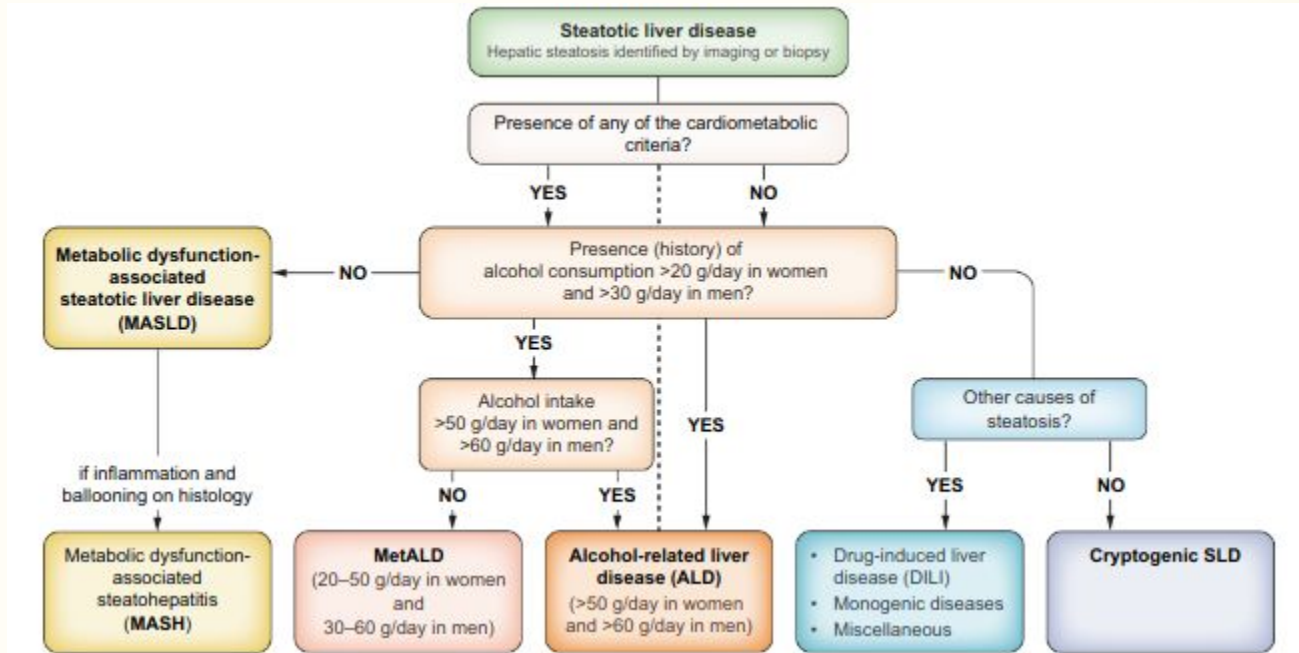
Bile Acid (BA) adalah molekul sinyal yang mengintegrasikan metabolisme lemak–glukosa–kolesterol di sumbu gut–liver–brain axis

Jam biologis (SCN & clock genes) mengatur ritme metabolik termasuk enzim sintesis BA; gangguan ritme (tidur/diet/alkohol) → “kacau” sinyal BA

Dysbiosis + leaky gut + inflamasi adalah jalur yang menghubungkan gangguan ritme/diet/alkohol ke obesitas–T2DM–fatty liver

Perubahan komposisi/BA pool oleh mikrobiota dapat mendorong hepatic steatosis melalui BA–FXR/FGF15/19

Flow Chart for SLD



Diagnosis Fatty Liver (MASLD)

Skrining awal – Cari steatosis hati (USG, biomarker, atau histologi) pada pasien:
Overweight/obesitas ($\text{BMI} \geq 25 \text{ kg/m}^2$, $\geq 23 \text{ kg/m}^2$ untuk Asia), T2DM, atau BMI normal + ≥ 2 kelainan metabolik

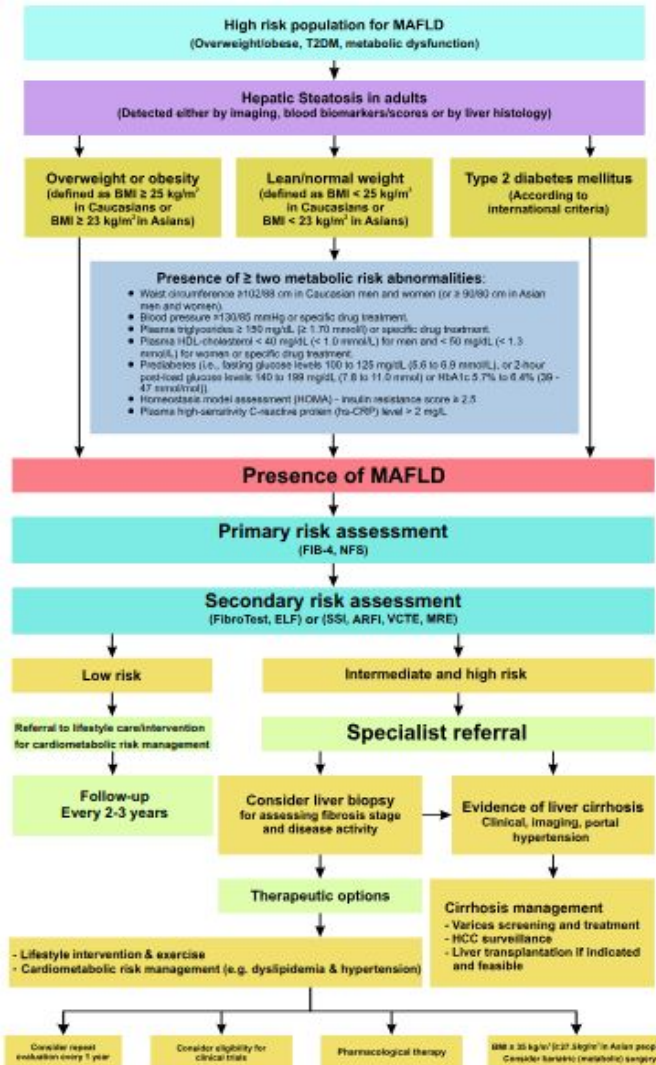
Tegakkan MAFLD – Steatosis + salah satu kondisi risiko di atas → diagnosis MAFLD

Primary risk assessment (fibrosis tahap 1) – Hitung FIB-4 / NFS

Secondary risk assessment (fibrosis tahap 2) – Bila nilai intermediate/tinggi, lanjutkan dengan FibroTest/ELF / elastografi (SSI, ARFI, VCTE, MRE)

Stratifikasi hasil

Low risk ($\text{FIB-4} < 1,3$) → edukasi & intervensi gaya hidup, monitor 2–3 th
Intermediate / High risk → lanjut ke langkah 6



Diagnosis Fatty Liver (MASLD)

Rujuk spesialis – FIB-4 $\geq 2,67$ / tes lanjutan tinggi → kirim ke hepatologi/gastroenterologi

Evaluasi di spesialis

Pertimbangkan biopsi hati bila diagnosis/derajat fibrosis masih meragukan atau sebelum terapi uji klinik
Konfirmasi sirosis jika ada

Sirosis terkonfirmasi – Kelola komplikasi (skrining varises & HCC, terapi portal hipertensi, evaluasi transplantasi bila perlu)

Intervensi sepanjang alur – Terapkan penurunan berat badan, diet-latihan terstruktur, & kontrol kardiometabolik (dislipidemia, hipertensi) sejak tahap awal

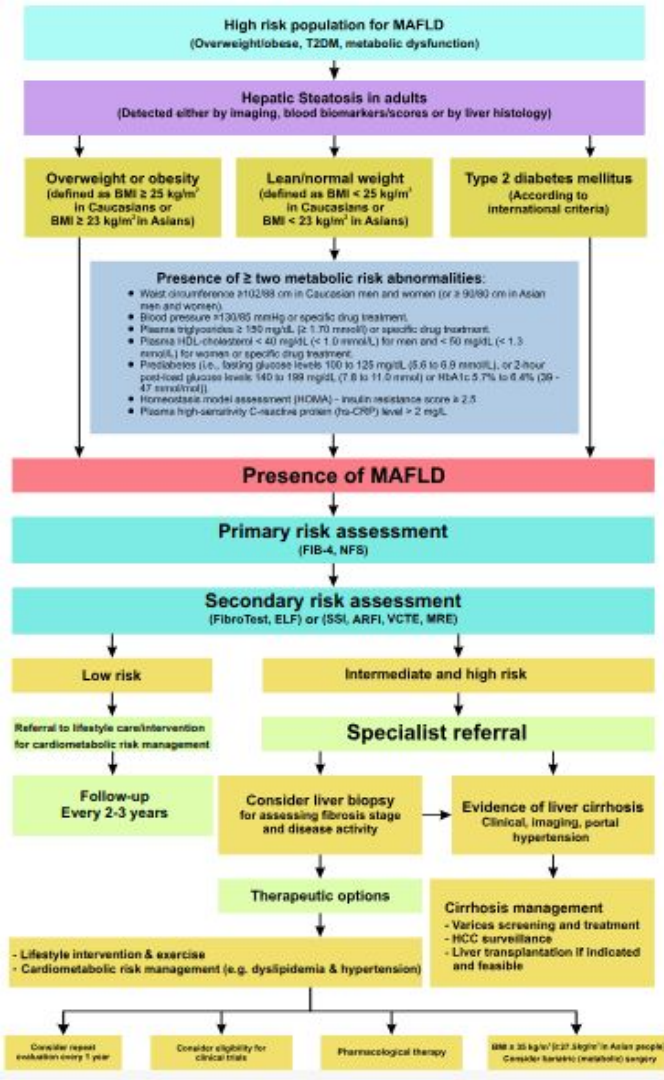


TABLE 1: List of drugs and agents as candidates for the treatment of NAFLD.

Insulin sensitizers
Metformin (Biguanide)
Thiazolidines (Pioglitazone, Rosiglitazone)
Antioxidants
Vitamin E
Vitamin C
Polyphenols (Resveratrol, etc.)
Hepatocyte-protective agents
Ursodeoxycholic acid
n-3 polyunsaturated fatty acids (EPA and DHA)
Antidyslipidemia
Statins
Fibrates
NPC1L1 inhibitors (ezetimibe)
RAS blockers
Angiotensin II receptor blockers
Angiotensin-converting enzyme inhibitors
Antialdosterone (spironolactone and eplerenone)
Renin inhibitor (aliskiren)
Incretin-related agents
GLP-1 agonists/analogs (exenatide and liraglutide)
DPP-4 inhibitors (sitagliptin and vildagliptin)
Antiobesity drugs
Orlistat
Sibutramine
Mazindol

Nakajima K. Multidisciplinary pharmacotherapeutic options for nonalcoholic Fatty liver disease. Int J Hepatol. 2012;2012:950693. doi:10.1155/2012/950693

Daftar Obat & Agen untuk Terapi Fatty Liver

UDCA sebagai hepatoprotective agent digunakan sebagai salah satu terapi untuk fatty liver


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graph TD
    ORLISTAT --> Obesity
    STATINS_EZETIMIBE[STATINS and EZETIMIBE] --> Dyslipidemia
    FXR_AGONISTS[FXR AGONISTS] --> Dyslipidemia
    LIFESTYLE_CHANGES[LIFESTYLE CHANGES] --> Obesity
    METFORMIN_TIAZOLIDINEDIONES[METFORMIN and TIAZOLIDINEDIONES] --> Insulin_resistance
    METFORMIN_TIAZOLIDINEDIONES --> Obesity
    GLP1_ANALOGS_DPP4_INHIBITORS[GLP-1 ANALOGS and DPP-4 INHIBITORS] --> Insulin_resistance
    CANNABINOID_CB2_ANTAGONISTS[CANNABINOID CB2 ANTAGONISTS] -.-> Obesity
    CANNABINOID_CB2_ANTAGONISTS -.-> NAFLD
    PENTOXIFYLLINE_ANTI_TNFα[PENTOXIFYLLINE and ANTI-TNFα] --> Pro_inflammatory_pathways
    RAS_BLOCKERS[RAS BLOCKERS] --> Pro_inflammatory_pathways
    PUFAS[PUFAs] --> Oxidative_stress
    PPAR_delta_PPAR_alpha_delta[PPAR δ PPAR α/δ] --> Oxidative_stress
    FIBRATES[FIBRATES] --> Oxidative_stress
    ANTIOXIDANTS[ANTIOXIDANTS] --> Oxidative_stress
    UDCA_DERIVATIVES[UDCA and DERIVATIVES] --> NASH
    MONOCLONAL_ANTIBODY[MONOCLONAL ANTIBODY] --> Fibrosis

    Insulin_resistance <--> Obesity
    Obesity <--> Dyslipidemia
    Insulin_resistance <--> NAFLD
    Obesity <--> NAFLD
    Dyslipidemia <--> NAFLD
    NAFLD --> NASH
    Oxidative_stress --> NASH
    Pro_inflammatory_pathways --> NASH
    NASH --> Fibrosis
    Fibrosis --> Cirrhosis_HCC[Cirrhosis HCC]
  
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The diagram illustrates the pathogenesis of Non-Alcoholic Fatty Liver Disease (NAFLD) and its progression to Non-Alcoholic Steatohepatitis (NASH), Fibrosis, and Cirrhosis/Hepatocellular Carcinoma (HCC). The central pathway shows the progression from NAFLD to NASH, which then leads to Fibrosis and finally Cirrhosis/HCC. Various metabolic and inflammatory pathways are shown, including Insulin resistance, Obesity, Dyslipidemia, and Pro-inflammatory pathways. Therapeutic interventions are indicated for each stage, such as GLP-1 analogs, Metformin, Lifestyle changes, FXR agonists, Statins, Ezetimibe, Orlistat, Cannabinoid CB2 antagonists, Pentyoxifylline, Anti-TNFα, RAS blockers, PUFAs, PPAR δ, PPAR α/δ, Fibrates, Antioxidants, UDCA and derivatives, and Monoclonal antibody.

02

UDCA

UDCA & Bile Acid

“Urso” dari Latin ursus = beruang; UDCA dominan pada empedu beruang

Asam empedu: steroid dari kolesterol, diproduksi hepatosit

Primer manusia: asam kolat (CA) & chenodeoksikolat (CDCA)

UDCA: epimer 7- β dari CDCA 1-3%
(komponen minor empedu manusia)

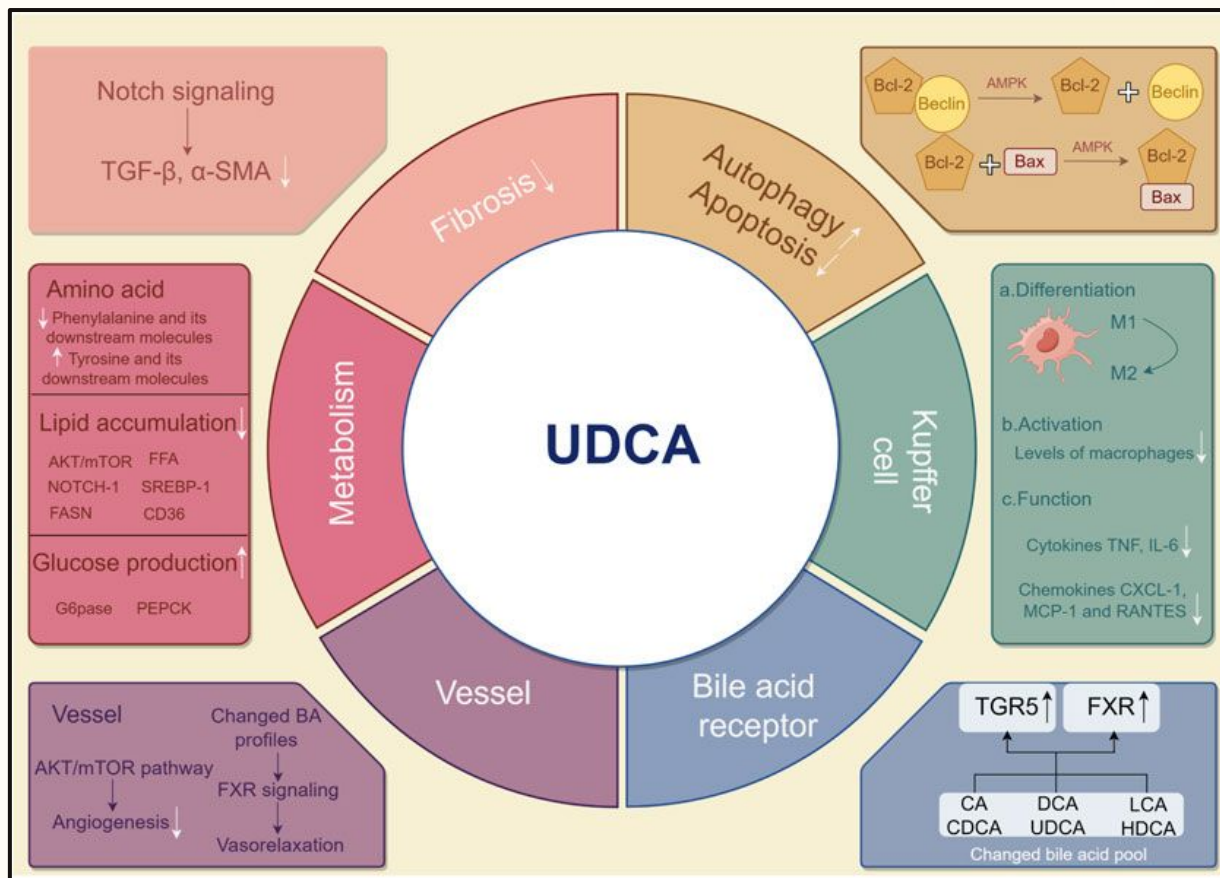
Farmakokinetik UDCA

Absorpsi 30–60% di usus halus

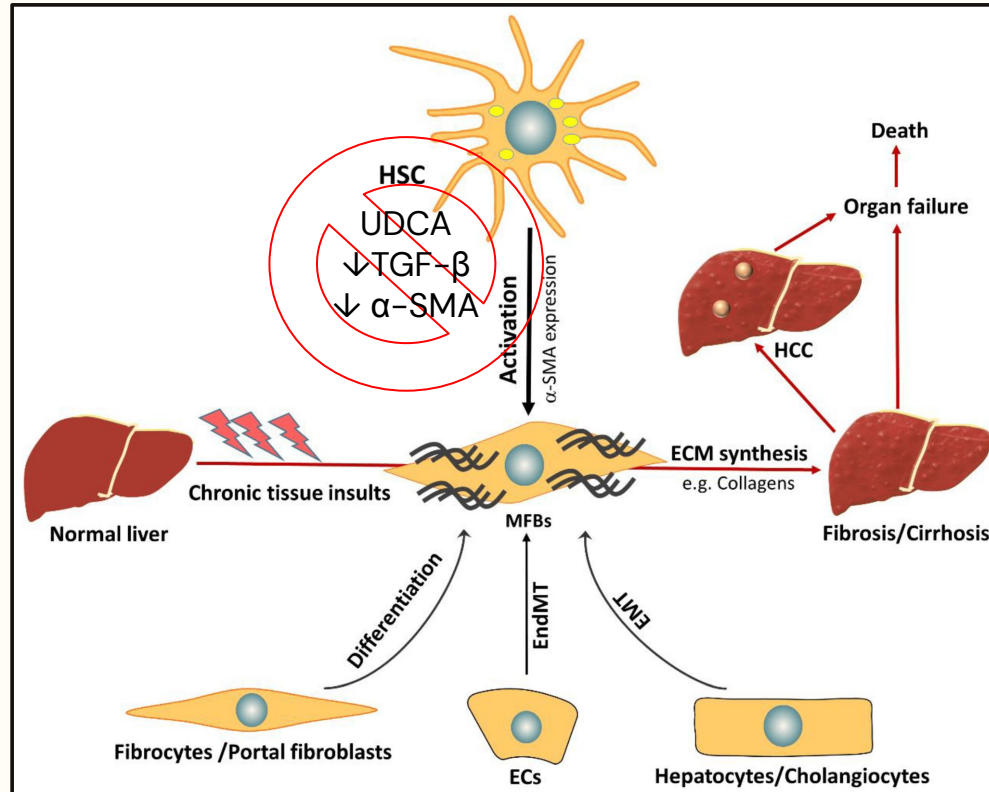
Dosis 13–15 mg/kg → UDCA 40–50% dari asam empedu bilier

First-pass extraction hepatic sekitar 50–70%

UDCA Mechanism

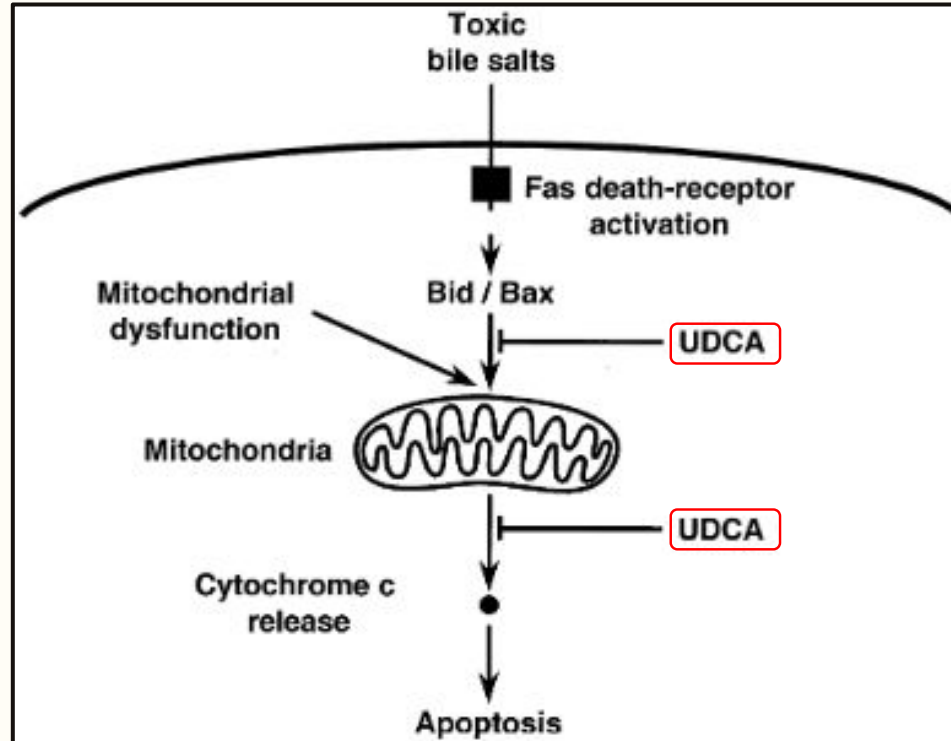


UDCA Mechanism



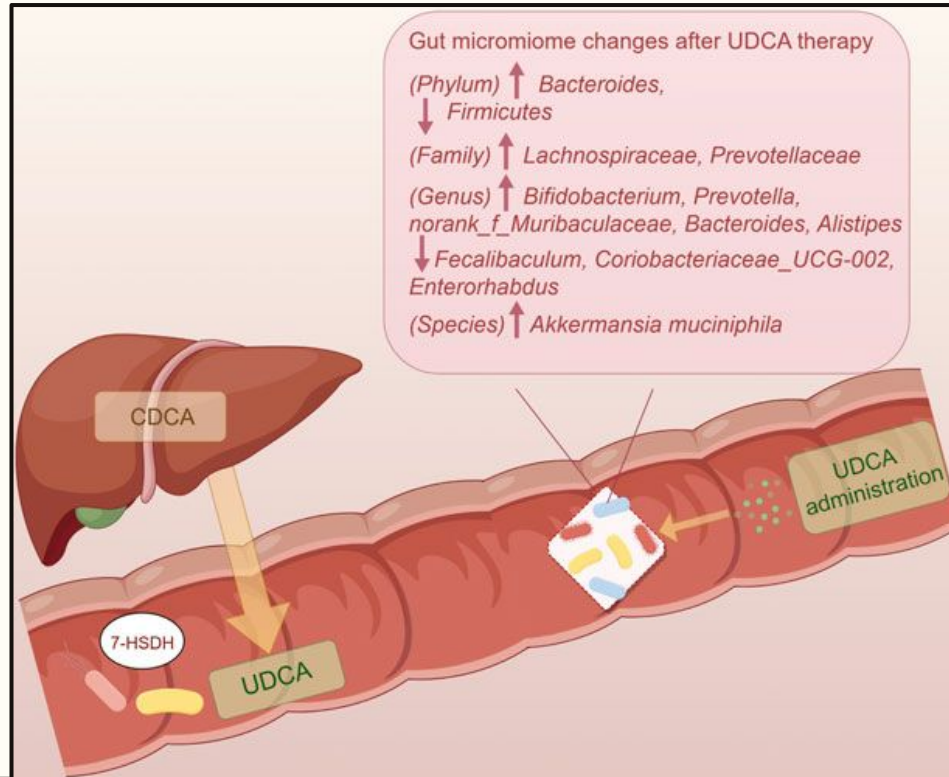
UDCA menghambat TGF- β → menurunkan ekspresi α -SMA

UDCA Mechanism



UDCA Mencegah disfungsi mitokondria & menghambat pelepasan sitokrom c

UDCA Mechanism



Pergeseran mikrobiota → kolam asam empedu lebih terkonjugasi → modulasi sinyal TGR5/FXR → homeostasis sumbu usus–hati (gut liver axis) membaik → disbiosis terkoreksi parsial

UDCA & Bile Acid Signaling in SLD

MASLD berkaitan dengan perubahan metabolisme kolesterol & bile acids serta sinyal metabolik hati–usus.

Modifikasi pool bile acids dapat memengaruhi:

- Stres hepatosit
- Inflamasi
- Jalur metabolik

REVIEW ARTICLE

Ursodeoxycholic Acid's Effectiveness in the Management of Nonalcoholic Fatty Liver Disease: A Systematic Review and Meta-analysis

Vaibhavi S Patel¹, Safa F Mahmood², Kunal H Bhatt³, Richisha M Khemkar⁴, Devanshi R Jariwala⁵, Bilal Harris⁶, Mirna M George⁷, Reuel A Kurudamannil⁸, Onyekachi E Anyagwa⁹, Rajeeva S Tak¹⁰, Maha Kassem¹¹

Received on: 08 April 2024; Accepted on: 10 May 2024; Published on: 12 June 2024

Metode:

Systematic review + meta-analysis 7 RCT NAFLD hingga 25 Juni 2023.

Hasil:

ALT turun signifikan ($p \leq 0,0001$)
AST turun signifikan ($p = 0,0009$)
GGT turun signifikan ($p \leq 0,0001$)

Table 1: Study characteristics

Author; year	Country	Duration	Dosage	Treatment	No. of participants at the start of the trial	No. of participants that completed the trial
Ratzu et al., 2011 ¹⁷	France	12 months	28–35 mg/kg/d	UDCA	60	55
				Placebo	66	61
Leuschner et al., 2010 ¹⁸	Germany, Greece	18 months	23–28 mg/kg/d	UDCA	95	78
				Placebo	91	82
Lindor et al., 2004 ¹⁹	United States, Canada	2 years	13–15 mg/kg/d	UDCA	78	55
				Placebo	86	64
Mojtahedi et al., 2023 ²⁰	Iran	3 months	5–10 mg/kg/d	UDCA	30	30
				Placebo	30	30
Elhini et al., 2022 ²¹	Egypt	6 months	5–10 mg/kg/d	UDCA	87	80
				Placebo	80	80
Santos et al., 2003 ²²	Brazil	3 months	10 mg/kg/d	UDCA	15	14
				Placebo	15	14
Méndez-Sánchez et al., 2004 ²³	Mexico	1.5 months	13–17 mg/kg/d	UDCA	14	12
				Placebo	13	11

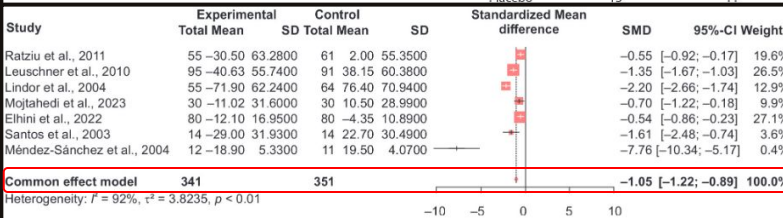


Fig. 3: Forest plot comparing the ALT values of intervention (UDCA) and control group

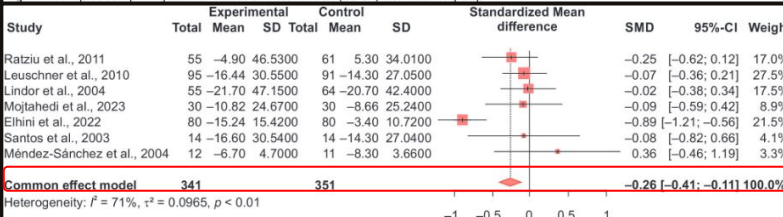


Fig. 4: Forest plot comparing the AST values of intervention (UDCA) and control group

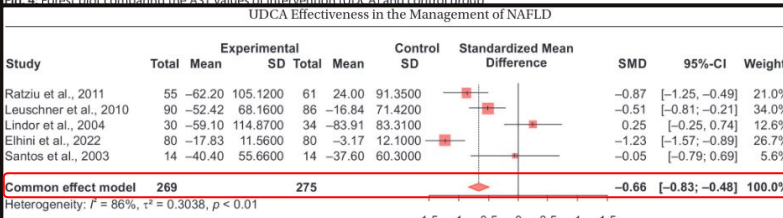


Fig. 5: Forest plot comparing the GGT values of intervention (UDCA) and control group

Clinical Trials Study

Ursodeoxycholic acid as a means of preventing atherosclerosis, steatosis and liver fibrosis in patients with nonalcoholic fatty liver disease

Metode:
Open-label, multicenter, international noncomparative trial
174 pasien NAFLD diberikan UDCA 15 mg/kg/hari selama 6 bulan + modifikasi gaya hidup

Hasil:
Penurunan signifikan pada enzim hati (ALT, AST, GGT) terutama dalam 3 bulan
Penurunan kadar kolesterol total, LDL, trigliserida, Indeks steatosis hati (FLI), ketebalan intima-media karotid (CIMT) secara signifikan

Parameter	Baseline	6 mo	P value
TC (total sample), mmol/L	6.03 ± 1.36	5.76 ± 1.21	< 0.001
Men	5.99 ± 1.39	5.77 ± 1.24	0.003
Women	6.11 ± 1.32	5.73 ± 1.16	0.019
LDL (total sample), mmol/L	3.86 ± 1.01	3.66 ± 0.91	< 0.001
Men	3.81 ± 1.04	3.67 ± 0.90	0.033
Women	4.0 ± 0.94	3.65 ± 0.93	0.006
HDL (total sample), mmol/L	1.24 ± 0.32	1.24 ± 0.27	0.910
Men	1.24 ± 0.33	1.21 ± 0.22	0.160
Women	1.23 ± 0.29	1.31 ± 0.34	0.036
TG (total sample), mmol/L	3.18 (2.00; 4.29)	2.04 (1.40; 3.16)	< 0.001
Men	3.13 (2.10; 4.29)	2.04 (1.4; 2.79)	< 0.001
Women	3.45 (1.80; 4.36)	2.26 (1.35; 3.62)	< 0.001
FLI (total sample)	84.9 ± 10.4	72.3 ± 17.6	< 0.001
Men	86.3 ± 9.0	73.6 ± 17.2	< 0.001
Women	81.9 ± 12.7	69.4 ± 18.4	< 0.001
FLI (total sample) ≥ 60/30-59/< 30	174 (100)/-/-	133 (76.4)/40 (23.0)/1 (0.6)	< 0.001
Men	121 (100)/-/-	96 (79)/24 (20)/1 (1)	< 0.001
Women	53 (100)/-/-	37 (70)/16 (30)/-	< 0.001
NFS (total sample) < -1.455 (F0-F2), n (%)	141 (81.0)	148 (85.0)	0.353
Men	102 (84.3)	112 (92.6)	0.071
Women	39 (76.6)	34 (64.2)	0.402
FIB 4 (total sample) < 1.45 (F0-F1), n (%)	157 (90.2)	163 (93.7)	0.256
Men	111 (91.7)	114 (94.2)	0.615
Women	46 (86.8)	49 (92.5)	0.402
CIMT (total sample), mm	0.985 ± 0.243	0.968 ± 0.237	0.013
Men	0.993 ± 0.224	0.977 ± 0.217	0.073
Women	0.967 ± 0.284	0.947 ± 0.276	0.052

Randomized Placebo-Controlled Trial of Ursodeoxycholic Acid With Vitamin E in Nonalcoholic Steatohepatitis

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Metode:

48 Pasien NASH terkonfirmasi biopsi & peningkatan enzim hati (AST dan ALT), dosis UDCA 12–15 mg/kg/hari 2 tahun

- 15 Kelompok 1: UDCA + vitamin E
- 18 Kelompok 2: UDCA + plasebo
- 15 Kelompok 3: Plasebo + plasebo

Hasil:

Kelompok UDCA/P menunjukkan penurunan ALT

Kelompok UDCA/Vit E menunjukkan penurunan signifikan kadar AST dan ALT (enzim hati)

Kesimpulan:

UDCA menurunkan kadar enzim hati ALT, dengan kombinasi Vitamin E dapat menurunkan AST & ALT serta memperbaiki steatosis hati pada pasien NASH selama 2 tahun.

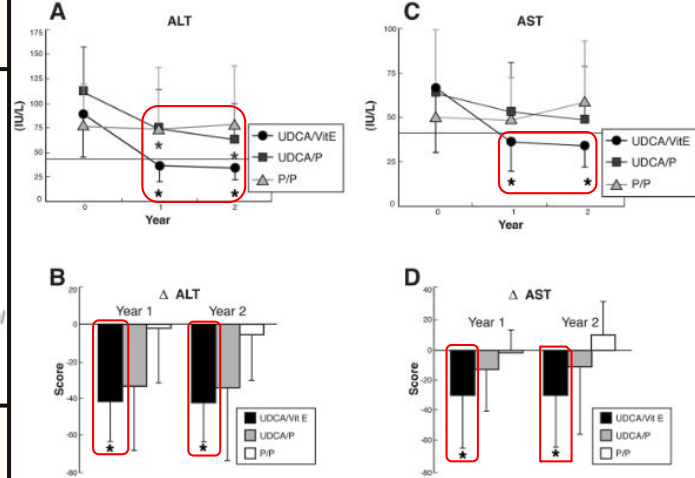


Figure 2. Effects on serum aminotransferase levels. (A) Evolution of ALT during the study period. Asterisks indicate a difference in comparison to value at inclusion ($P < .05$). The horizontal line represents the upper normal limit. (B) Changes from baseline in ALT levels at 1 year and at end of the study. Asterisks indicate a difference in comparison to the changes measured in the P/P group ($P < .05$). (C) Evolution of AST during the study period. Asterisks indicate a difference in comparison to value at inclusion ($P < .05$). The horizontal line represents the upper normal limit. (D) Changes from baseline in AST levels at 1 year and at end of the study. Asterisks indicate difference in comparison to the changes measured in the P/P group ($P < .05$). Values represent means $\pm$ standard deviations.

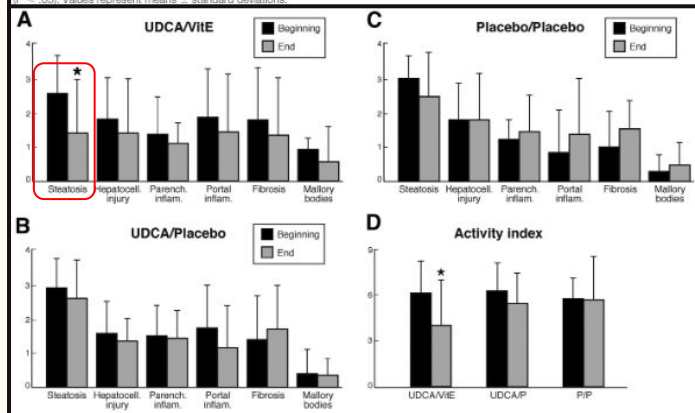


Figure 3. Effects on liver histology. (A) Scores for 6 histologic features of nonalcoholic steatosis (steatosis, hepatocellular injury, parenchymal inflammation, portal inflammation, fibrosis, and presence of Mallory bodies) at entry in the study and at the end of the 2 years of treatment with UDCA in combination with vitamin E. Asterisks indicate a significant difference in comparison to value at inclusion. (B) Scores for the biopsies of the patients who received UDCA and placebo. (C) Scores for the biopsies of the patients who received placebo and placebo. (D) Changes in the activity index, which represents steatosis, hepatocellular injury, and parenchymal inflammation, at entry and at completion of the study for the 3 groups. Asterisks indicate a significant change in the combination group. Values are means $\pm$ standard deviations.

Impact of combined therapy with alpha-lipoic and ursodeoxycolic acid on nonalcoholic fatty liver disease: double-blind, randomized clinical trial of efficacy and safety

Vincenzo Gianturco · Giovanni Troisi · Alessia Bellomo · Scialla Bernardini · Elisa D'Ottavio · Valeria Formosa · Cristina Lo Iacono · Walter Verrusio · Benedetta Marigliano · Vincenzo Marigliano

Metode:

- RCT double-blind, plasebo-kontrol, 4 kelompok, durasi 12 bulan
- Asam alfa-lipoat (ALA) 400 mg/hari + UDCA 300 mg/hari vs ALA saja, UDCA saja, atau diet (plasebo)

Hasil:

Penurunan signifikan ALT ($p \leq 0,0001$), AST ($p = 0,0009$), dan GGT ($p \leq 0,0001$) pada kombinasi ALA+UDCA dibanding tiga kelompok lainnya.

Kesimpulan:

Kombinasi ALA 400 mg/hari + UDCA 300 mg/hari selama 12 bulan menurunkan ALT/AST/SGT

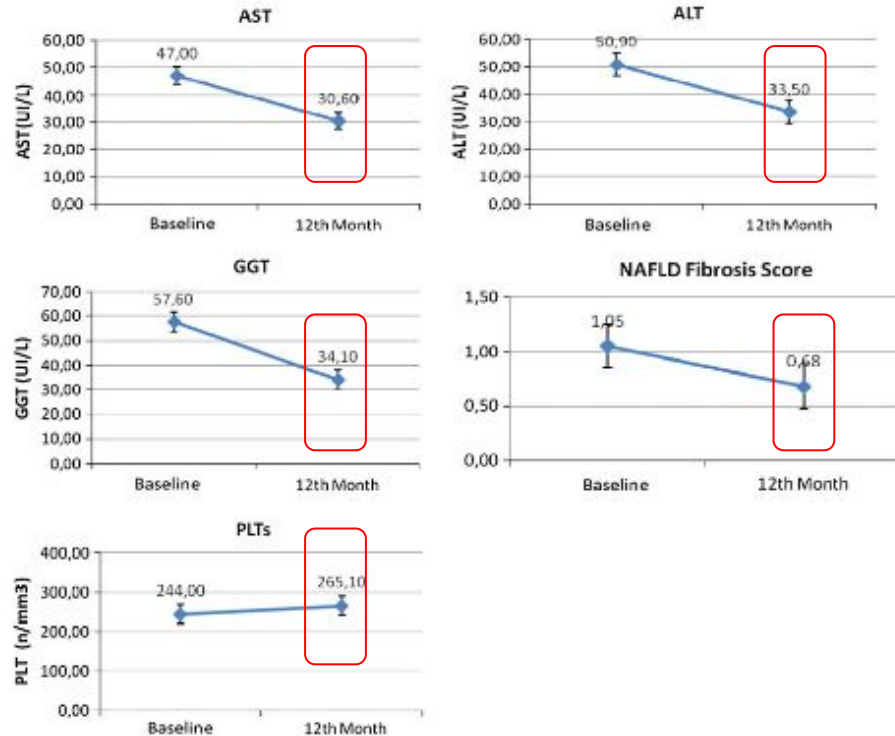


Fig. 2 Comparison between baseline and 12th month's evaluation in ALAUDCA group. This figure shows baseline and 12th month evaluation of ALAUDCA group. Graphs underline the significant

improvements in AST, ALT, GGT, PLTs, and NAFLD fibrosis score values after 12 months of treatment (values are expressed as mean $\pm$ SD)

Table 3 Post-hoc evaluation with Bonferroni method

12 months evaluation	ALAUDCA Mean (SD)	ALA Mean (SD)	Bonferroni p value	UDCA Mean (SD)	Bonferroni p value	PLACEBO Mean (SD)	Bonferroni p value
AST (U/L)	30.6 (3)	49.2 (3.2)	0.0001	48.9 (3.5)	0.0003	49.6 (3.3)	0.0001
ALT (U/L)	33.5 (3.2)	52 (3.5)	0.0003	51.8 (3.3)	0.0004	52.9 (3.9)	0.0002
GGT (U/L)	34.1 (3.6)	59.1 (3.7)	0.0007	59.4 (3.7)	0.0006	59.7 (3.8)	0.0004
NAFLD fibrosis score	0.68 (0.1)	1 (0.3)	0.004	1.14 (0.2)	0.003	1.02 (0.2)	0.003
PLTs (n/mm ³)	265.1 (16)	230 (29)	0.005	226 (19)	0.002	233.7 (19)	0.005

META-ANALYSIS

Efficiency of ursodeoxycholic acid for the treatment of nonalcoholic steatohepatitis: A systematic review and meta-analysis

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Metode:

Systematic review + meta-analysis 8 RCT NAFLD hingga Januari 2022

Hasil:

ALT turun signifikan ($p = 0,012$)

GGT turun signifikan ($p \leq 0,00$)

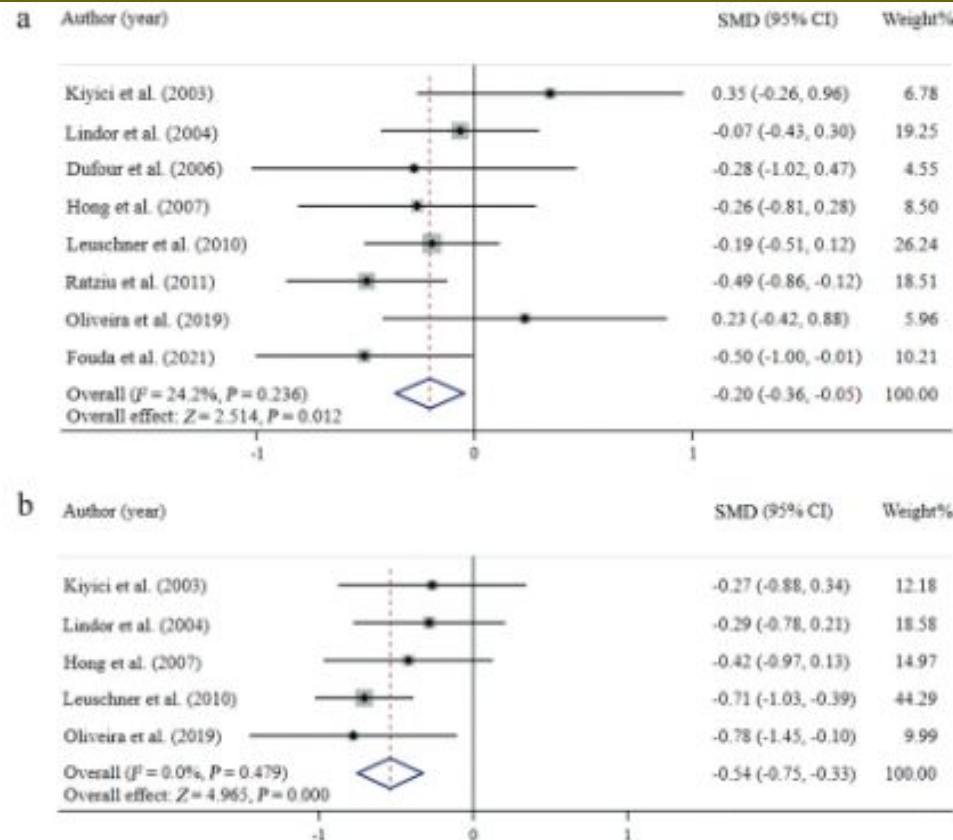
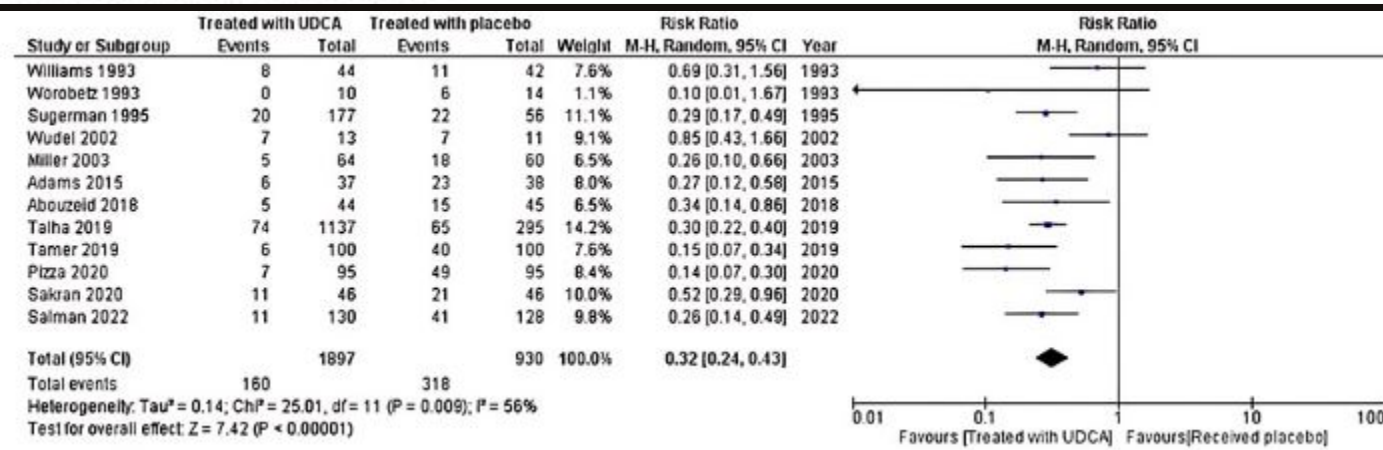


Figure 3. Forest plots of the effect of UDCA on serum ALT (a) and GGT levels (b).

Ursodeoxycholic Acid Prophylaxis and the Reduction of Gallstone Formation After Bariatric Surgery: An Updated Meta-Analysis of Randomized Controlled Trials

Mohammad Al-hunuti¹, Yousef Alsardia¹, Alaa Odeh¹, Belal Bdour¹, Ramadan Hassanat¹, Ali Aloun¹, Ban W. Sha'ban¹, Sara M. Nseirat¹



Metode:
Meta-analisis RCT (1993–2022),
n=2 767, 12 studi

Hasil:
UDCA turunkan risiko total: RR
0,13; p<0,0001

Subkelompok: 3, 6, 12
bulan—semua lebih rendah
Simptomatik & kolesistektomi
juga berkurang signifikan

Ursodeoxycholic acid in the management of symptomatic gallstone disease: systematic review and clinician survey

Lewis Hall¹, James Halle-Smith², Richard Evans³, Giles Toogood⁴, Tom Wiggins⁵ , Sheraz R Markar⁶, Spyros Kapoulas⁵, Paul Super⁵, Olga Tucker⁵ and Siobhan C McKay^{2,3,*}

Metode:

Tinjauan sistematis: 4 RCT, 3 prospektif, 1 retrospektif

Hasil:

7/8 studi mendukung UDCA mengurangi nyeri bilier

Table 3 Summary table of studies included in systematic review									
Patient: adults with symptomatic gallstone disease; Setting: hospital, outpatients; Intervention: UDCA treatment; Comparison: usual care or placebo where possible; Outcome: biliary pain/colic episodes									
Author (Country)	Design	(1) No. of patients (2) Gallstone cohort (3) Intervention (4) Comparator	Age (years), mean(s.d.)	Sex ratio (M:F)	(1) Follow-up (2) UDCA dose	Outcome and results	Opinion on UDCA	Quality of the evidence (GRADE)	Comments
Venneman (Netherlands) ¹²	RCT	177 (5G) (I) UDCA (89 of 177) (C) Placebo (88 of 177)	47(1) 45(1)	1.4 3.7	77 days (7-244) 750mg/d	Colic episodes in 6 months 11 of 81 (14%); 6 of 81 (7%)	Unfavourable	Moderate	Short treatment duration
Polli (Italy) ¹³	RCT	42 (5G) (I) UDCA (23 of 42) (C) Placebo (19 of 42)	Adults	1:3.2	14 days 300mg/d	Dyspeptic symptoms at 0, 7, and 14 days (1) Severity (19 patients): mean score reduced 6.00 to 2.11 to 0.94 (2) Episodes (23 patients): average score reduced 5.61 to 2.57 to 1.39 (1) Severity (19 patients): 5.84 to 4.05 to 3.47 (2) Episodes (16 patients): 5.75 to 4.33 to 3.06	Favourable	Low	Very short treatment duration
Frigerio (Italy) ¹⁴	RCT	240 (5G) (I) UDCA (C) Placebo	Adults		14 days 300 mg/d	(1) Reduction dyspeptic symptoms (2) Reduction pain symptoms (1) of 120 (53%); (2) 82 of 101 (81%); (1) 45 of 120 (54%); (2) 47 of 99 (47%)	Favourable	Moderate	Very short treatment duration
Petroni (UK/Italy) ¹⁵	RCT*	154 (5G) (I) UDCA	46.9(1.7)	23:56	2 years 30mg/kg/d	Pain episodes 3 months pre- and post-treatment 59% versus 26% (P<0.001)	Favourable	High	Single-arm assessed Satisfactory treatment duration
Tomida (Japan) ¹⁶	PC	(C)* pre- versus post-treatment 527 (5G + ASG) (I) UDCA (181 of 527) (74 of 181 5G)	55.4(13.0)	2:3	66.4(40) 600mg/d	Reduced risk pain at 10-years 63%	Favourable	Low	+ASG patients. Satisfactory treatment duration
Tini (USA) ¹⁷	PC†	(C) No UDCA (146 of 527) (112 of 546 5G) 11 (5G + ASG) (I) UDCA	54.3(12.8)	2:3	6-38 months 250-300 mg/d	92% (P<0.001, RR 0.19, 95% CI 0.10-0.34) Pre-treatment colic episodes 22 of 53 (41.5%), 6 months post-treatment 5 of 53 (5.7%) (not reported by dose)	Favourable	Low	+ASG patients Variable treatment duration
Polli (Italy) ¹⁸	PC‡	116 (5G + ASG) (I) UDCA (78 of 116)	51	4:9	6-12 months 5-6/10-12 mg/kg/d	Improvement in dyspeptic/pain episodes 85% pts improvement	Favourable	Low	+ASG patients Single-arm assessed Moderate treatment duration
Meredith (UK) ¹⁹	RC§	(C)‡ pre- versus post-treatment 88 (5G + ASG) (I) UDCA (46 of 88) (C)§ pre- versus post-treatment	53.2(2.2)	1:2.5	18 months 2.5-10 mg/kg/d	Colic episodes pre- versus during treatment 72% reduction	Favourable	Low	+ASG patients Single-arm assessed Satisfactory treatment duration

Take Home Messages

UDCA merupakan asam empedu hidrofilik yang menetralkan efek toksik bile acid dan melindungi hepatosit dari kerusakan.

UDCA memperbaiki fungsi hati melalui efek anti-inflamasi, anti-apoptosis, dan anti-fibrotik pada sel hati.

Meta-analisis RCT terbaru menunjukkan UDCA menurunkan kadar ALT, AST, dan GGT secara signifikan pada pasien fatty liver disease.

Kombinasi UDCA dengan agen antioksidan seperti asam alfa-lipoat atau vitamin E menunjukkan hasil biokimia yang lebih baik dibandingkan monoterapi.

UDCA efektif sebagai terapi adjuvan pada MASLD/SLD bila dikombinasikan dengan modifikasi gaya hidup untuk hasil optimal.

Terima Kasih